



# Excel Formula Masterclass

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## 1. SUM Function

Purpose: Adds numbers together.

**Syntax**:

`=SUM(number1, [number2], ...)`

|   | A        | B      | C | D |
|---|----------|--------|---|---|
| 1 | Product  | Amount |   |   |
| 2 | Pen      | 50     |   |   |
| 3 | Book     | 200    |   |   |
| 4 | Bag      | 500    |   |   |
| 5 | Notebook | 150    |   |   |
| 6 |          | 900    |   |   |

*Annotations* → `=SUM(B2:B5)`

**When to use SUM?**

- \* Total sales
- \* Total expenses
- \* Total marks
- \* Monthly totals

**Tip**: Use **SUM** instead of adding many cells manually.



## 2. AVERAGE Function

- Calculates the arithmetic mean of numbers.



### Syntax:

= AVERAGE (number1, [number2], ...)

### Example Table

|   | A       | B     |
|---|---------|-------|
| 1 | Student | Marks |
| 2 | Rahul   | 80    |
| 3 | Anita   | 75    |
| 4 | Vikram  | 90    |
| 5 | Sara    | 85    |
| 6 |         |       |

Range: B2:B5

### Formula

= AVERAGE (B2:B5)

Takes the values  
from cells B2 to B5

### Result

B6 82.5

Average of 80, 75, 90, 85  
is 82.5

### How it works

$$\begin{array}{ccccc} \boxed{80 + 75 + 90 + 85} & = & \boxed{\frac{330}{4}} & = & \boxed{82.5} \\ \text{Total of values} & & \text{Number of values} & & \text{Average} \end{array}$$

Average = Total ÷ Number of Values

### Common Use Cases

- Marks of students
- Average salary of employees
- Average sales performance
- Customer ratings / feedback



Use **AVERAGE** to quickly find the typical or average value from a set of numbers.



**Tip:** **AVERAGE** ignores blank cells and text values.



## 3. COUNT Function

- Counts cells containing numbers.



### Syntax:

= COUNT(value1, [value2], ...)

### Example Table

|   | A      | B     |
|---|--------|-------|
| 1 | Name   | Marks |
| 2 | Rahul  | 80    |
| 3 | Anita  | 75    |
| 4 | Vikram | -     |
| 5 | Sara   | 90    |
| 6 |        |       |

Range: B2:B5

### Formula

= COUNT(B2:B5)

Counts how many cells contain numbers in B2 to B5.

### Result

|    |   |
|----|---|
| B6 | 3 |
|----|---|

There are 3 cells with numbers.

### What gets counted vs. what doesn't

#### ✓ COUNTED (Numbers)

| Cell | Value | Counted? |
|------|-------|----------|
| B2   | 80    | Yes ✓    |
| B3   | 75    | Yes ✓    |
| B5   | 90    | Yes ✓    |

Numbers (including 0, negative numbers, decimals) are counted.

#### ✗ NOT COUNTED

| Cell | Value   | Counted? |
|------|---------|----------|
| B4   | -       | No ✗     |
| B6   | (blank) | No ✗     |
| A2   | Rahul   | No ✗     |
| A3   | Anita   | No ✗     |

Text values and blank cells are **NOT** counted.



**Tip:** Use **COUNT** when you only want to count cells that contain numbers and ignore text or blanks.



## 4. COUNTA Function

- Counts non-empty cells.



Syntax:

= COUNTA(value1, [value2], ...)

Example Table

|   | A      |
|---|--------|
| 1 | Name   |
| 2 | Rahul  |
| 3 | Anita  |
| 4 | Vikram |
| 5 |        |
| 6 | Sara   |

Range: A2:A6

Formula

= COUNTA(A2:A6)

Counts all non-empty cells in the range A2 to A6.

Result

|    |   |
|----|---|
| A7 | 4 |
|----|---|

There are 4 non-empty cells in A2:A6.



COUNTA counts text + numbers + other non-empty values. It counts any cell that is not blank.

What COUNTA counts:

| Example Value | Type    | Counted? | Remarks                     |
|---------------|---------|----------|-----------------------------|
| Rahul         | Text    | ✓ Yes    | Text values are counted     |
| 123           | Number  | ✓ Yes    | Number values are counted   |
| 01-01-2025    | Date    | ✓ Yes    | Dates are counted           |
| TRUE          | Logical | ✓ Yes    | Logical values are counted  |
| 0             | Number  | ✓ Yes    | Zero is counted             |
| (blank cell)  | Blank   | ✗ No     | Blank cells are not counted |



Tip: Use COUNTA when you want to count all filled cells, regardless of what is inside them.



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## ④ COUNTA Function

→ Counts non-empty cells.

Syntax:

=COUNTA(value1, [value2], ...)

Example:

|   | A            |
|---|--------------|
| 1 | Name         |
| 2 | Aryan        |
| 3 | Ayan         |
| 4 | Aryan Kelvin |
| 5 |              |
| 6 | Sara         |

Non-empty cells

Blank cell

Formula:

=COUNTA(A2:A6)

|   | A            |
|---|--------------|
| 1 | Name         |
| 2 | Aryan        |
| 3 | Ayan         |
| 4 | Aryan Kelvin |
| 5 |              |
| 6 | Sara         |

Result:

|                             |   |
|-----------------------------|---|
| Non-empty Cells<br>(Result) | 4 |
|-----------------------------|---|

↑  
There are 4 non-empty  
cells in the range.

★ COUNTA counts  
text + numbers +  
other non-empty values.

★ Note: COUNTA counts any cell that is not empty.  
It includes text, numbers, dates, logical values, errors, etc.



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## 5 COUNTBLANK Function

→ Counts empty cells in a range.

Syntax:

=COUNTBLANK(range)

Example:

|   | A        | B      |
|---|----------|--------|
| 1 | Employee | Status |
| 2 | Rahul    | Done   |
| 3 | Anita    |        |
| 4 | Vikram   | Done   |
| 5 | Sara     |        |
| 6 | Imran    | Done   |

← Blank

← Blank



COUNTBLANK counts cells that are truly empty.

Formula:

=COUNTBLANK(B2:B6)

|   | A        | B      |
|---|----------|--------|
| 1 | Employee | Status |
| 2 | Rahul    | Done   |
| 3 | Anita    |        |
| 4 | Vikram   | Done   |
| 5 | Sara     |        |
| 6 | Imran    | Done   |

Result:

|                      |   |
|----------------------|---|
| Empty Cells (Result) | 2 |
|----------------------|---|

↑  
There are 2 blank (empty) cells.



Useful for finding missing data.

★ Note: COUNTBLANK counts only truly empty cells.  
Cells with spaces (" ") or formulas returning "" are not counted.



## ⑥ MAX Function

→ Returns the largest number.

Syntax:

=MAX(number1, [number2], ...)

Example:

|   | A     |
|---|-------|
| 1 | Sales |
| 2 | 1200  |
| 3 | 2500  |
| 4 | 1800  |
| 5 | 4500  |
| 6 | 3200  |

← Largest Value



MAX returns the highest number in a range.

Formula:

=MAX(A2:A6)

|   | A     |
|---|-------|
| 1 | Sales |
| 2 | 1200  |
| 3 | 2500  |
| 4 | 1800  |
| 5 | 4500  |
| 6 | 3200  |

Result:

|                         |      |
|-------------------------|------|
| Largest Number (Result) | 4500 |
|-------------------------|------|

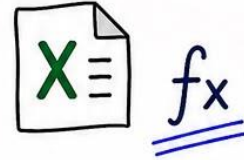
↑  
This is the largest number in the range.

★ Note: MAX works only with numbers.  
Text values are ignored.



## 7 MIN Function

→ Returns the smallest number.



Syntax:

=MIN(number1, [number2], ...)

### Example Table

|   | A     |
|---|-------|
| 1 | Sales |
| 2 | 1200  |
| 3 | 2500  |
| 4 | 1800  |
| 5 | 4500  |
| 6 | 3200  |

← Smallest Value

### Formula

=MIN(A2:A6)

→ Returns the smallest number in the range A2 to A6.

### Result

A7 1200

The smallest number in the range A2:A6 is 1200.



MIN returns the **smallest numeric value** in a range or list.  
Text values and blanks are ignored.

### What MIN considers:

| Example Value | Type    | Considered? | Remarks                      |
|---------------|---------|-------------|------------------------------|
| 1200          | Number  | ✓ Yes       | Numbers are considered       |
| 2500          | Number  | ✓ Yes       | Numbers are considered       |
| Rahul         | Text    | ✗ No        | Text values are ignored      |
| 01-01-2025    | Date    | ✓ Yes       | Dates are treated as numbers |
| TRUE          | Logical | ✓ Yes       | Logical TRUE is treated as 1 |
| 0             | Number  | ✓ Yes       | Zero is considered           |
| (blank cell)  | Blank   | ✗ No        | Blank cells are ignored      |



**Tip:** Use MIN when you want to find the lowest value (minimum) in a range of numbers.



## 8. IF Function

→ Returns one value if a condition is TRUE and another if FALSE.



Syntax: =IF(logical\_test, value\_if\_true, value\_if\_false)

### Example Table

|   | A       | B     | C      |
|---|---------|-------|--------|
| 1 | Student | Marks | Result |
| 2 | Rahul   | 75    |        |
| 3 | Anita   | 42    |        |
| 4 | Vikram  | 88    |        |

We will apply IF formula in C2 and copy down.

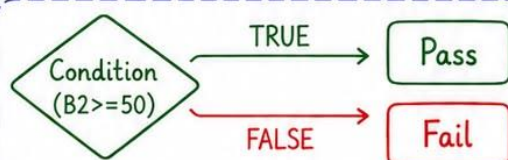
### Formula

=IF(B2>=50, "Pass", "Fail")

Checks if Marks (B2) is greater than or equal to 50.

### Result

|   | A       | B     | C      |
|---|---------|-------|--------|
| 1 | Student | Marks | Result |
| 2 | Rahul   | 75    | Pass   |
| 3 | Anita   | 42    | Fail   |
| 4 | Vikram  | 88    | Pass   |



IF checks the condition.  
If TRUE → returns "Pass"  
If FALSE → returns "Fail"



IF function evaluates a condition and returns:

- value\_if\_true → when the condition is TRUE
- value\_if\_false → when the condition is FALSE

### How IF Function Works:

| Example (Marks in B) | Logical Test | Result (Displayed) | Why?                              |
|----------------------|--------------|--------------------|-----------------------------------|
| 75                   | 75 >= 50     | Pass               | 75 is greater than or equal to 50 |
| 42                   | 42 >= 50     | Fail               | 42 is less than 50                |
| 88                   | 88 >= 50     | Pass               | 88 is greater than or equal to 50 |



Tip: IF is one of the most important Excel functions.  
It is the foundation of many powerful formulas.



## 9. SUMIF Function



→ Adds values that meet one condition.

Syntax : =SUMIF(range, criteria, [sum\_range])

### Example Table

|   | A        | B          | C     |
|---|----------|------------|-------|
| 1 | Employee | Department | Sales |
| 2 | Rahul    | Sales      | 5000  |
| 3 | Anita    | HR         | 3000  |
| 4 | Vikram   | Sales      | 7000  |
| 5 | Sara     | HR         | 4000  |

Only "Sales" rows are considered.

### Formula

=SUMIF(B2:B5, "Sales", C2:C5)

→ Adds sales values where department is "Sales" (in B2:B5).

### Result

Total Sales 12000

Sales department total is 12000.



SUMIF adds values based on a single condition.

You can use it with both the same column or a different range.

### What SUMIF Considers:

| Example Value | Type   | Considered? | Remarks                       |
|---------------|--------|-------------|-------------------------------|
| 5000          | Number | ✓ Yes       | Numbers are considered        |
| Sales         | Text   | ✓ Yes       | Text (criteria) is considered |
| HR            | Text   | ✓ Yes       | Text (criteria) is considered |
| 0             | Number | ✓ Yes       | Zero is considered            |
| (blank cell)  | Blank  | ✗ No        | Blank cells are ignored       |



Tip: Use SUMIF when you want to add values based on one specific condition.



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## 10. SUMIFS Function

→ Adds values that meet multiple conditions.



Syntax:

=SUMIFS(sum\_range, criteria\_range1, criteria1, ...)

### Example Table

|   | A        | B          | C     | D     |
|---|----------|------------|-------|-------|
| 1 | Employee | Department | Month | Sales |
| 2 | Rahul    | Sales      | Jan   | 5000  |
| 3 | Anita    | Sales      | Feb   | 7000  |
| 4 | Vikram   | HR         | Jan   | 3000  |
| 5 | Sara     | Sales      | Jan   | 6000  |

Department = Sales

Month = Jan

(Only rows 2 and 5 match both conditions.)

### Formula

=SUMIFS(D2:D5, B2:B5, "Sales", C2:C5, "Jan")

→ Adds sales for Department = Sales and Month = Jan.

### Result

Total Sales **11000**

Sales for Sales department in Jan is 11000.



SUMIFS works with multiple conditions.

All conditions must be met for a row to be included in the sum.

### What SUMIFS Considers:

| Example Value | Type   | Considered? | Remarks                            |
|---------------|--------|-------------|------------------------------------|
| 5000          | Number | ✓ Yes       | Number (sales value) is considered |
| Sales         | Text   | ✓ Yes       | Text (department) is considered    |
| Jan           | Text   | ✓ Yes       | Text (month) is considered         |
| 0             | Number | ✓ Yes       | Zero is considered                 |
| (blank cell)  | Blank  | ✗ No        | Blank cells are ignored            |



**Tip:** Use SUMIFS when you need to add values based on multiple conditions (e.g., department + month, region + product, etc.).



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## 11. COUNTIF Function

→ Counts cells that meet one condition.



Syntax: =COUNTIF(range, criteria)

### Example Table

|   | A        | B         |
|---|----------|-----------|
| 1 | Employee | Status    |
| 2 | Rahul    | Completed |
| 3 | Anita    | Pending   |
| 4 | Vikram   | Completed |
| 5 | Sara     | Completed |
| 6 | Imran    | Pending   |

Only  
"Completed"  
cells are  
counted.

### Formula

=COUNTIF(B2:B6, "Completed")

Counts how many cells  
in B2:B6 are "Completed".

### Result

Result 3

There are 3 employees  
with "Completed" status.



COUNTIF counts cells based on a single condition.

It can be used with text, numbers or dates.

### What COUNTIF Considers / Ignores:

| Example Value | Type   | Considered? | Remarks                     |
|---------------|--------|-------------|-----------------------------|
| Completed     | Text   | ✓ Yes       | Text is counted             |
| Pending       | Text   | ✓ Yes       | Text is counted             |
| 123           | Number | ✓ Yes       | Number is counted           |
| 01-01-2025    | Date   | ✓ Yes       | Date is counted             |
| (blank cell)  | Blank  | ✗ No        | Blank cells are not counted |



Tip: Use COUNTIF when you want to count cells  
that meet a specific condition.